

Two-arms cGAS-dependence asymmetry across methods

IFN/ISG arm (top) is cGAS-dependent: significant, positive Interaction. HIF/glycolysis arm (bottom)
rises in BOTH heat arms but is flat in the Interaction - no detectable cGAS-dependence at n=5.
[PROVISIONAL - inferred sample mapping (Hspa1b/Hsph1 thermometer + Cgas), pending collaborator sample sheet; n=5/group]

		WT heat (39 vs 37 °C)	cGAS-KO heat (39 vs 37 °C)	Interaction	Heat main (pooled)
IFN / ISG arm (cGAS-dependent)	PROGENy JAK-STAT	-6.5	-11.7	+16.8	-9.3
	DE:limma-trend Irf7	+0.9	-9.3	+7.2	-6.0
	DE:limma-trend Oasl2	-0.8	-10.6	+6.9	-8.0
	DE:limma-trend Ifit1	+4.2	-4.3	+6.0	-0.1
	TF:CollecTRI Irf3	-1.3	-3.2	+5.9	-2.3
	TF:CollecTRI Stat2	-0.8	-2.6	+5.8	-1.7
	DE:limma-trend Mx1	+5.7	-2.5	+5.8	+2.2
	TF:CollecTRI Stat1	+2.0	+0.3	+5.3	+1.1
	DE:limma-trend Isg15	+4.0	-2.4	+4.5	+1.2
	TF:CollecTRI Irf1	-0.6	-1.7	+3.4	-1.2
	GSEA:Hallmark Interferon Alpha Response	-1.9	-2.5	+3.2	-2.2
	GSEA:Reactome Interferon Alpha Beta Signaling	-1.5	-1.8	+2.9	-1.7
	GSEA:Hallmark Interferon Gamma Response	-1.8	-2.2	+2.9	-2.0
	GSEA:GO_BP Antiviral Innate Immune Response	+1.3	-1.2	+2.7	-0.9
	GSEA:WikiPathways Type I Interferon Induction And Signaling During Sarscov2 Infection	-1.3	-2.0	+2.6	-1.7
	GSEA:GO_BP Type I Interferon Mediated Signaling Pathway	-1.3	-1.7	+2.6	-1.6
	GSEA:GO_BP Interferon Mediated Signaling Pathway	-1.5	-1.8	+2.5	-1.6
	GSEA:Reactome Interferon Signaling	-1.4	-1.7	+2.5	-1.6
	GSEA:GO_BP Response To Type I Interferon	-1.4	-1.8	+2.4	-1.7
	GSEA:WikiPathways Hostpathogen Interaction Of Human Coronaviruses Interferon Induction	-1.3	-1.7	+2.4	-1.6
HIF / glycolysis arm (no detectable cGAS-dependence at n=5)	GSEA:WikiPathways Angiogenesis Overview	+0.9	-0.9	+1.8	+0.9
	GSEA:Reactome Regulation Of Pyruvate Metabolism	-1.5	-1.9	+1.7	-1.7
	GSEA:Reactome Signaling By Vegf	+1.5	+1.5	+1.4	+1.5
	GSEA:KEGG Vegf Signaling Pathway	+1.7	+1.5	+1.4	+1.6
	GSEA:GO_BP Regulation Of Glucose Metabolic Process	-1.3	-1.5	+1.3	-1.4
	GSEA:WikiPathways Angiogenesis	+1.0	+0.9	+1.2	+0.9
	GSEA:Reactome Pyruvate Metabolism	-1.2	-1.3	+1.2	-1.3
	GSEA:WikiPathways Vegfavegfr2 Signaling	+2.0	+1.9	+1.2	+1.9
	GSEA:GO_BP Glucose Metabolic Process	+1.2	+1.1	+1.1	+1.1
	GSEA:MitoXplorer Pyruvate Metabolism	-1.4	-1.4	+1.1	-1.4
	GSEA:Reactome Vegfr2 Mediated Vascular Permeability	+0.9	+1.0	+1.1	+1.0
	GSEA:GO_BP Pyruvate Metabolic Process	+0.9	+1.0	+0.9	+1.0
	GSEA:GO_BP Regulation Of Cell Migration Involved In Sprouting Angiogenesis	+1.6	+1.5	-0.9	+1.5
	GSEA:GO_BP Glycolytic Process Through Glucose 6 Phosphate	+1.1	+1.2	-1.1	+1.2
	GSEA:MitoXplorer Glycolysis	+1.3	+1.4	-1.1	+1.3
	GSEA:KEGG Pyruvate Metabolism	-1.3	-1.2	-1.1	-1.2
	GSEA:Hallmark Glycolysis	+1.6	+1.6	-1.1	+1.6
	GSEA:Hallmark Hypoxia	+1.9	+2.0	-1.3	+2.0
	GSEA:KEGG Glycolysis Gluconeogenesis	+1.0	+1.1	-1.3	+1.0
	PROGENy Hypoxia	+4.0	+4.6	-2.2	+4.4

Tile = signed score; orange = up in numerator, blue = down. Black ring = padj < 0.05. Claim tier: L3 (provisional, n=5/group). 'Flat Interaction' is NOT proven cGAS-indep